

YASH KUMAR

Full Stack Developer | AI-Augmented Builder | Competitive Programmer

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About

First-year undergraduate pursuing B.S. in Data Science and AI at IIT Madras and Computer Science & AI at Scaler School of Technology. Open-source contributor (GSSoC 2026) with experience building scalable full-stack applications. Passionate about software engineering, competitive programming, and leveraging AI coding agents (Claude, Cursor) as a force multiplier for high-velocity development and shipping production-grade code.

Education

B.S. – Data Science and AI <i>Indian Institute of Technology, Madras</i>	2025 – 2029
CS and AI <i>Scaler School of Technology, Bengaluru</i>	2025 – 2029

Technical Skills

Programming Languages	Python, Java, JavaScript, Dart, TypeScript
Web & App Development	HTML, CSS, React.js, Node.js, Flask, Vite, Flutter, Firebase, Authentication, Firestore, REST APIs
Databases	MongoDB (Basic), MySQL, PostgreSQL, Supabase
AI & LLM Tools	Cursor, Claude, Groq API (LLaMA 3.3), Hugging Face APIs
Data and Libraries	NumPy, Pandas, Matplotlib, SciPy, Seaborn
Tools & Platforms	Git, GitHub, VS Code, Linux, Docker, Jupyter Notebook
Core Concepts	Data Structures & Algorithms, OOP, DBMS, OS
Soft Skills	Problem Solving, Team Collaboration, Communication, Adaptability

Projects

RepoOwl AI-Powered GitHub Issue Triage Extension [Repo Link] <i>TypeScript, React, Supabase (PostgreSQL/RLS), Groq (LLaMA 3.3), Flutter Web, Dart, Chrome Extension API</i>	- Architected a fully client-side Chrome Extension for real-time semantic duplicate detection on GitHub repositories using the Groq API (LLaMA 3.3), bypassing heavy backend requirements to keep token costs at zero. - Designed a role-aware “Dual-Layer” database architecture (Maintainer Hub & Contributor Sandbox) using Supabase, implementing secure PostgreSQL Row Level Security (RLS) policies. - Utilized AI coding agents (Cursor, Claude) as a force multiplier to learn Dart and independently build/ship a responsive, CodeRabbit-inspired production landing page in Flutter Web in a 48-hour sprint. - Managed complex background worker queues to handle asynchronous GitHub API fetching, LLM rate-limiting, and local Chrome storage caching.
AlgoGATE Competitive Programming Platform [Repo Link] <i>React.js, Vite, Tailwind CSS, Firebase, Firestore, Codeforces API, Highlight.js, Vercel</i>	- Built a structured competitive programming platform to guide users from CF-800 to CF-1500 with topic-wise progression - Replaced tag-based filtering with topic-wise + rating gradient problem sequencing - Integrated Codeforces API for real-time problem data and user progress synchronization - Implemented personalized problem recommendations based on user progress and solved history - Developed notes and real-time discussion section powered with Firestore - Built a real-time sync system with manual refresh control for updated submissions and stats - Optimized frontend performance using Vite + modular React architecture for faster load times - Deployed scalable full-stack application using Firebase (Auth + Firestore) and Vercel

E-Commerce Website (Frontend) [\[Repo Link\]](#)

React.js, Vite, Tailwind CSS, Context API, Axios, DummyJSON API, React Hook Form, Yup, Formspree

- Developed a responsive e-commerce frontend with product browsing, filtering, wishlist, and full cart management system
- Implemented global state management using Context API and optimized performance with custom hooks (debounced search, reusable logic)
- Integrated APIs and built a complete checkout flow with automated order notifications via Formspree

Yulu Demand Analysis | Hypothesis Testing & Regression [\[Repo Link\]](#)

Python, Pandas, NumPy, Matplotlib, Seaborn, SciPy, Scikit-learn

- Performed EDA and statistical hypothesis testing (t-tests, chi-square) on 10K+ records to identify key drivers of micro-mobility demand
- Built a linear regression model ($R^2 \approx 35\%$) to quantify feature impact, identifying hour, temperature, and humidity as major predictors
- Derived actionable insights such as weather-based fleet optimization and peak-hour demand patterns, enabling data-driven business decisions

ClaimLens – Scientific Claim Verification Environment | Meta OpenCV Hackathon [\[Repo Link\]](#)

FastAPI, React.js, Vite, Tailwind CSS, Docker, Hugging Face APIs

- Developed an OpenEnv-compliant scientific claim verification platform for evaluating AI agent reasoning and factual accuracy
- Built a FastAPI backend for claim processing, evaluation pipelines, grading logic, and API services
- Designed a React + Vite frontend for interactive experimentation, benchmarking, and result visualization
- Integrated Hugging Face LLM APIs to automate scientific claim verification and reasoning workflows
- Containerized the application using Docker and Docker Compose for reproducible deployment and testing
- Implemented structured evaluation mechanisms to benchmark AI agent performance across verification scenarios

Deepfake Video Detection MVP | Auraverse 2.0 [\[Repo Link\]](#)

Vision Transformer (ViT), DINOv2, PyTorch, OpenCV, Streamlit

- Developed a deepfake video detection system using a self-trained Vision Transformer (ViT-B/14) with a DINOv2 backbone for detecting manipulated video content
- Designed a temporal analysis pipeline using frame extraction, confidence scoring, median smoothing, and segment merging to identify suspicious video intervals
- Implemented timestamp-level localization of manipulated segments, enabling fine-grained detection beyond binary video classification
- Optimized inference efficiency through 2 FPS frame sampling while maintaining reliable temporal coverage across videos
- Built an interactive Streamlit interface for video upload, visualization of detected segments, and confidence-based deepfake analysis
- Evaluated model performance using video-level and segment-level predictions on deepfake benchmark datasets

Co-Curricular Activities

Member, Open-Source Club

2025 – Present

- Participated in hackathons, contests, and tech workshops within the department
- Collaborated on open-source mini-projects and contributed to knowledge-sharing sessions

Competitive Programming Enthusiast – Codeforces, Member: The NlogN Club

Online | 2026 – Present

- Solved 200+ problems on Codeforces in Java and Python
- Max Rating – 930 | Max Streak – 30 days
- Codeforces Profile: codeforces.com/profile/GODMODFURY

Other Activities

- Participated in Oratory Club events, including a **mock venture capital (VC) pitch**
- Actively participate in and complete goals through college fitness challenges.

Achievements

- Completed 200 problems milestone on Codeforces
- Participated in AuraVerse 2.0, Meta OpenCV, and SIH Hackathon
- Selected for InVariant and Impulse CF contest finals in ASCENT event by Scaler
- Selected as an Open-Source contributor at GirlScript Summer of Code (GSSoC 2026)
- Successfully built and maintained an open-source repository, **RepoOwl**, with active issue and pull request (PR) management for **SWOC 2026**.

Languages

English – Professional Proficiency | Hindi – Native